

Dear Selection Committee,

This letter is to express my motivation in the PhD position advertised at VM, NTNU. My experiences reflect my curiosity and strong personal drive to understand natural history through a genomics lens for the benefit of biodiversity and society.

Growing up as the youngest in a modest family affected by violence, I was fortunate to be able to study and pursue my passion for science. Over time, my gratitude evolved into a strong sense of responsibility, driven by my awareness of Peru's immense potential and pressing needs. Early in my studies, I joined research groups at my university, where scarce resources encouraged creativity and close collaboration. Working with diverse people and learning new methods allowed me to refine my interests, while also building strong technical abilities and interpersonal skills.

Under the mentorship of Prof. Gisella Orjeda, during my BSc, I led a DNA barcoding study on Antarctic lichens during the pandemics, where despite difficult challenges, I earned the highest honors. Furthermore, we developed an interdisciplinary project on algarrobo, a keystone species native to the Peruvian coastal desert whose decline threatens rural families. My leadership in multiple steps led to the first chromosome-scale genome of a wild species in Peru.

I was recently given the exciting opportunity of joining NTNU for my master's studies under Prof. Michael Martin and Dr. Vanessa Bieker, as part of the BiGTREE network. This has been my most enriching experience, as delving into ancient DNA I also sparked my interest in ancient civilizations, resulting in significant discoveries regarding algarrobo's genetic diversity. In addition to my thesis project, Dr. Sarah Martin's exciting ideas motivated me to explore sedimentary ancient DNA as a unique tool for archaeologists to reconstruct past histories, resulting in VM's first sedaDNA study.

Given the rich diversity and history of Mexico and its parallels in Peru, both cradles of civilization in the Americas, I am convinced that the support of co-supervisor Dr. Marcela Sandoval will contribute significantly to understanding ecological change in the Neotropics, establishing DNA as an effective tool for paleoecological explorations. NTNU's VM offers an ideal environment for this research, with my supervisors' expertise and its inspiring surroundings.

As evidenced by my experience and collaborations with VM scientists, I am confident that my background, technical skills, and commitment to interdisciplinary research will contribute meaningfully to this project and to NTNU's mission.

Thank you for your consideration.

Sincerely,

Renato La Torre